

MileStone-2

Project Title

AI-Based Medical Symptom Analysis & Disease Prediction System

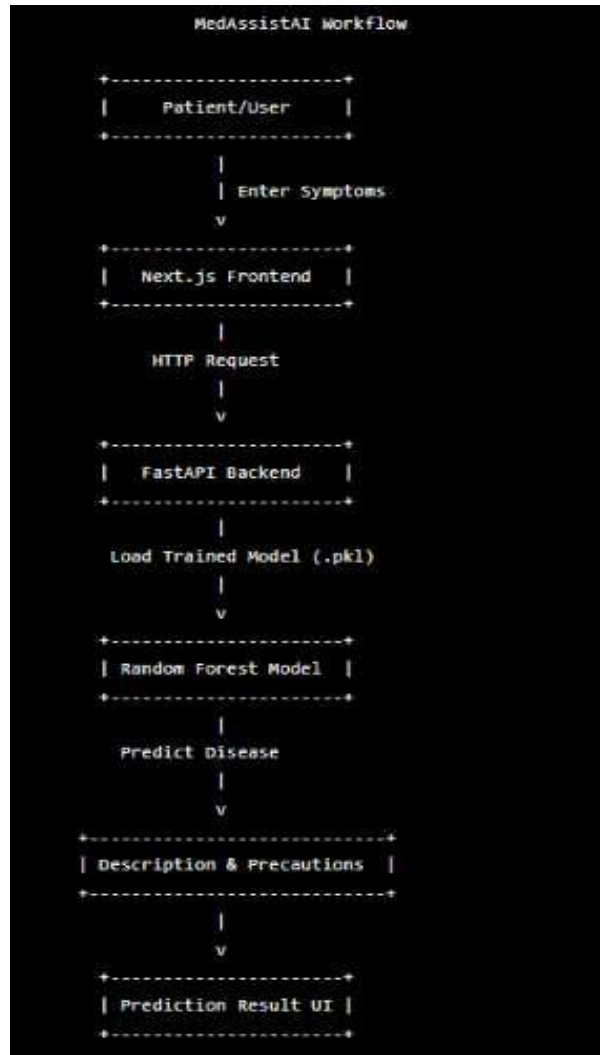
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1. Project Description

The MedAssistAI system is an end-to-end Artificial Intelligence based healthcare application designed to predict possible diseases using patient symptoms. The system utilizes machine learning algorithms to analyze symptom patterns, demographic information, and patient inputs to provide disease predictions along with confidence scores, disease descriptions, and recommended precautions.

The project combines a Random Forest Classification Model with a FastAPI backend and a Next.js frontend to create an intelligent disease prediction platform. The backend exposes REST APIs for disease prediction, while the frontend provides an interactive interface for patients and healthcare professionals. The platform is further designed to maintain patient prediction history, generate medical reports, and provide a dedicated dashboard for doctors to review patient records.



2. Dataset Used

The MedAssistAI project has been upgraded by replacing the previous merged symptom dataset with a larger and more comprehensive disease-symptom dataset obtained from Kaggle.

Dataset Source:

[Disease and Symptoms Dataset \(Kaggle\)](#)

- Approximately **246,000** patient records
- **773** disease categories
- **377** symptom features

Dataset Processing

Before model development, Exploratory Data Analysis (EDA) was performed to understand the quality of the dataset.

Shape: (246945, 378), **diseases:** 773, **duplicates:** 57298

```
pd.Series(df['diseases']).value_counts().describe()
```

```
count    773.000000
mean     319.463131
std       351.691549
min        1.000000
25%       30.000000
50%      168.000000
75%      505.000000
max     1219.000000
```

Name: count, dtype: float64

The following preprocessing steps were carried out:

- Removed duplicate patient records
- Verified disease labels
- Checked missing values
- Analyzed disease distribution
- Verified symptom columns
- Examined feature consistency

After removing duplicate records:

Final Dataset Shape

- **Rows:** 189,647
- **Columns:** 378 (377 Symptoms + Disease Label)

Disease Selection

Although the dataset contains **773 diseases**, many diseases contained only a very small number of samples, creating a severe class imbalance problem.

To build a reliable classification model:

- Initially selected diseases with sufficient medical relevance.
- Applied a minimum sample threshold of **90 samples per disease**.
- Final dataset contained **90 well-represented disease classes** for model training.

This filtering significantly reduced class imbalance and improved model performance.

```
count      773.000000
mean       245.338939
std        333.650389
min         1.000000
25%        10.000000
50%        77.000000
75%       315.000000
max      1219.000000
Name: count, dtype: float64
Diseases: 773
Dataset shape: (189647, 378)
```

Original dataset

```
count      90.000000
mean       674.411111
std        342.463150
min        92.000000
25%       409.250000
50%       794.000000
75%       825.000000
max      1215.000000
Name: count, dtype: float64
Remaining diseases: 90
Dataset shape: (60697, 378)
```

Filtered dataset

3. Data Exploration

Before training the machine learning models, detailed Exploratory Data Analysis (EDA) was performed to understand the dataset characteristics.

Analysis Performed

- Dataset structure analysis
- Missing value analysis
- Duplicate record detection
- Disease frequency analysis
- Symptom frequency analysis
- Class imbalance visualization
- Distribution of diseases
- Correlation between symptom features

The analysis showed that the dataset was highly imbalanced, with many diseases containing only a few patient records. This motivated the disease filtering strategy adopted before model training.

```
df.columns
Index(['diseases', 'anxiety and nervousness', 'depression',
      'shortness of breath', 'depressive or psychotic symptoms',
      'sharp chest pain', 'dizziness', 'insomnia',
      'abnormal involuntary movements', 'chest tightness',
      ...,
      'stuttering or stammering', 'problems with orgasm', 'nose deformity',
      'lump over jaw', 'sore in nose', 'hip weakness', 'back swelling',
      'ankle stiffness or tightness', 'ankle weakness', 'neck weakness'],
      dtype='str', length=378)
```

```
diseases:90, shape:(60697, 378)
null:0, duplicates:0
```

4. Data Preprocessing

Several preprocessing techniques were applied before model training.

4.1 Duplicate Removal

Duplicate patient records were identified and removed to eliminate redundant information.

```
raw dataset
shape:(246945, 378), duplicates:57298
```

```
processed dataset
shape:(189647, 378), duplicates:0
```

4.2 Disease Filtering

Only diseases having at least **90 samples** were retained for training.

Final Disease Count:

90 Diseases

```
processed dataset
Diseases: 773
Dataset shape: (189647, 378)
```

```
filtered dataset
Diseases: 90
Dataset shape: (60697, 378)
```

4.3 Feature Selection

Selected all **377 symptom features** as input variables.

4.4 Label Encoding

Disease names were encoded into numerical labels for supervised learning.

count	60697.000000										
mean	42.723215										
std	26.321584										
min	0.000000										
25%	19.000000										
50%	43.000000										
75%	67.000000										
max	89.000000										
dtype:	float64										

...	problems with orgasm	nose deformity	lump over jaw	sore in nose	hip weakness	back swelling	ankle stiffness or tightness	ankle weakness	neck weakness	target
...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	54.0
...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	64.0
...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	59.0
...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	31.0
...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	38.0

4.5 Train-Test Split

The processed dataset was divided into training and testing datasets using stratified sampling to preserve disease distribution.

```
X_train:(48557, 377), y_train:(48557,)
X_test:(12140, 377), y_test:(12140,)
```

5. Model Training and Evaluation

Multiple machine learning and deep learning approaches were implemented and compared to identify the most suitable model for disease prediction.

Models Implemented

Traditional Machine Learning

- Random Forest
- Decision Tree
- Logistic Regression
- Support Vector Machine
- K-Nearest Neighbors

Ensemble Learning

- Voting Classifier (SOFT VOTING ENSEMBLE)

```
=====
MODEL PERFORMANCE
=====
Random Forest Accuracy : 0.8968
XGBoost Accuracy      : 0.8745
ExtraTrees Accuracy   : 0.8759
=====

Ensemble Weights
{'rf': 0.3442461935122319, 'xgb': 0.3273831594812655, 'et': 0.32837064700650254}
```

- Stacking Classifier

```
=====
Random Forest Accuracy : 0.8969
=====
Extra Trees Accuracy : 0.8773
=====
LightGBM Accuracy : 0.8842
=====
CatBoost Accuracy : 0.9040
=====
```

```
=====
Stacking ( Meta)
=====
Accuracy : 0.8992
=====
```

- Boosting Models

- XGBoost

```
XGBoost Accuracy : 0.8818
XGBoost F1 Score  : 0.8820
```

- LightGBM

```
LightGBM Accuracy : 0.8772
LightGBM F1 Score  : 0.8771
```

- CatBoost

```
Catboost Accuracy : 0.9016
Catboost F1 Score  : 0.9011
```

- Extra Trees

```
Extra Trees Accuracy : 0.8759
Extra Trees F1 Score  : 0.8815
```

- Random Forest

```
Random Forest Accuracy : 0.8968
Random Forest F1 Score  : 0.8998
```

Deep Learning

- Artificial Neural Network (ANN) (Early stopping)

```
Epoch 16/150
81/81 ————— 2s 23ms/step Test Accuracy: 0.9009884595870972
0.2137 - learning_rate: 3.6450e-07
```

Pre-trained Tabular Model

- TabNet (Early stopping)

```
Accuracy: 0.8942
```

```
Early stopping occurred at epoch 55 with best_epoch = 35 and best_Validation_accuracy = 0.89415
```

Each model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion matrix

6. Model Comparison

The performance of all trained models was compared to identify the most accurate model.

Model	Technique	Accuracy
Random Forest	Boosting	89.6%
XGBoost	Boosting	88.1%
Extra Trees	Linear	87.5%
Voting Classifier	Ensemble	88.6%

Model	Technique	Accuracy
Stacking Classifier	Ensemble	89.9%
ANN	Deep Learning	90.0%
TabNet	Pre-trained Deep Tabular Model	89.4%
CatBoost	Gradient Boosting	90.1%

7. Conclusion

The second milestone focused on improving the disease prediction model by replacing the previous dataset with a significantly larger and more diverse disease-symptom dataset.

Key achievements include:

- Replaced the previous dataset with a Kaggle disease-symptom dataset.
- Performed comprehensive Exploratory Data Analysis (EDA).
- Cleaned and filtered the dataset to address class imbalance.
- Reduced the dataset to **90 well-represented diseases** for reliable model training.
- Implemented and evaluated multiple machine learning, ensemble learning, deep learning, and pre-trained tabular models.
- Achieved the highest accuracy of **90.1% using CatBoost**, followed by **90.0% using ANN**.
- Selected CatBoost as the final prediction model for integration into the MedAssistAI application.